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MyDesign Portfolio

Element D Initial Template

How to use this template

Teachers: You may customize this document before providing it to students.

Students: The template below is one way to present your design ideas, but the specific assignment given by your instructor may require a different structure or dictate different details. If you make a copy of this template to edit for your own design ideas, be sure to delete the instructions.

Guidance before filling out the template on the next page:

Because you will iterate on your design multiple times, you will need to create multiple copies of these sections. Think of Element D as a diary that you will date and keep adding to throughout your work until your final iteration. It's critical that iteration shines through in this Element.

You should begin with many design ideas that reflect the individual ideas of your group members and then the group ideas. Number these ideas and include a date, description, drawing, and details for each.

After you have created an initial set of ideas, document your decision making on which design to include, ideally with a design matrix. Create a plan of work, ideally using a Gantt chart that may be housed in this document or in a different day-to-day document. Generate a design blueprint with enough details that someone else not familiar with your project could also build your design.

You will then proceed forward to build and test this design, returning to Element D afterwards when you are ready to create more design ideas. Fill out each of these steps again.

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Element D: Design Concept Generation, Analysis, and Selection

Design Idea Generation (D1)

This section focuses on the process of generating solutions. The initial rounds should include every idea you come up with but the subsequent rounds of ideation should lead to viable options. Each idea should include not only words, but also a picture. Use as many copies of this sub-template as needed.

Design Idea #1

Date: 2/9/26

Description: We could create a google presentation with 15- 19 slides that we can do for the whole school or go to each class and present this.

Drawing that includes distinguishing information such as scale, dimensions, and materials:



Design Idea #2

Date: 2/9/26

Description: We could check the nozzles on the machine to see whether or not they're tightly screwed in cause it probably might be the reason for the problem

Drawing that includes distinguishing information such as scale, dimensions, and materials:

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Design Idea #3

Date: 2/9/26

Description: We could check the level of carbon dioxide and syrup the fountain is giving to see if there is too much or too little, measuring the perfect amount.

Drawing that includes distinguishing information such as scale, dimensions, and materials:



Design Idea Comparison and Selection (D1, D2)

When you've finished generating ideas, you must compare your design options according to all your design requirements. Incorporate the priority of those requirements when comparing design options. A decision matrix (perhaps a Pugh Matrix) is encouraged since it allows for systematic, weighted comparisons. If using weights in a decision matrix, be sure to explain and justify the weighting based on your design requirements. You may use the weighted design matrix template below, inserting your criteria and design options.

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	weight	Scales, Dimensions, Materials	Checking Nozzles	Fixing level of Carbon Dioxide	Option 4
Signal	20%	5	5	5	
Can contact the people in the lunch room	15%	1	2	2	
Very High Tech Device	20%	2	5	5	
Cost	30%	1	3	3	
Accuracy	10%	3	4	5	
Keeping it Clean	5%	5	3	3	
score					

Design Blueprint (D4)

Describe all of your design's details such that someone unfamiliar with your project could build your prototype. Include sketches, dimensions, CAD (if taught in your class), manufacturing methods, material choices, supply lists, and costs.

Carbon Dioxide and Syrup Levels: Making a Sensor

Date: 2/9/26

Sketch:

Manufacturing Methods: Repetitive because we want consistent results and accurate results from our sensor.

Step 1: First the sensor must have a target variable or else it wouldn't work well at all because without a true assignment it makes the sensor useless.

Step 2: Build a circuit to detect when the syrup level is low to let the sensor know so that the sensor can inform the people in the lunch room.

Step 3: Lastly, a device that can always detect the ratio of syrup and water to make sure it stays balanced.

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Supply List and Costs:

Item	Cost each	# needed	Total
Circuit	570-1,000	1	570-1,000
Wiring	15-50	3	45-150
C02 Detector	10-200	1	10-200
Total Cost	595-1,250	5	625-1,350

Plan of Action (D3)

A Gantt chart is encouraged to lay out the primary steps expected on a visual timeline. Larger tasks from the Gantt chart should be broken down further into detailed sub-tasks, which may be housed in a different day-to-day document. Discrete tasks should include enough detail such that others could replicate the efforts, such as procedures, materials, tools, status, etc.

1. We first will program the sensor to be able to know when exactly the syrup to water ratio is low
2. After we do that we will use the wires to connect the sensors to the circuit which will allow the two things to connect and work together. The circuit will always be able to inform the sensor when exactly the syrup levels are low
3. Then we will also connect our C02 detector so that they can also make sure the carbon dioxide is okay because that is another important component in the soda machine
4. Then we will make the sensor be able to contact the people in the lunch room so they can increase the amount of syrup in the machine